

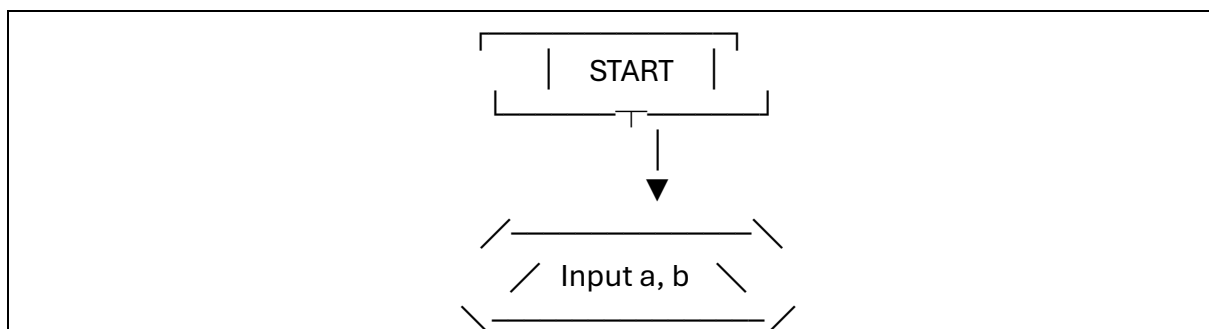
HDOC Day-9 Assignment

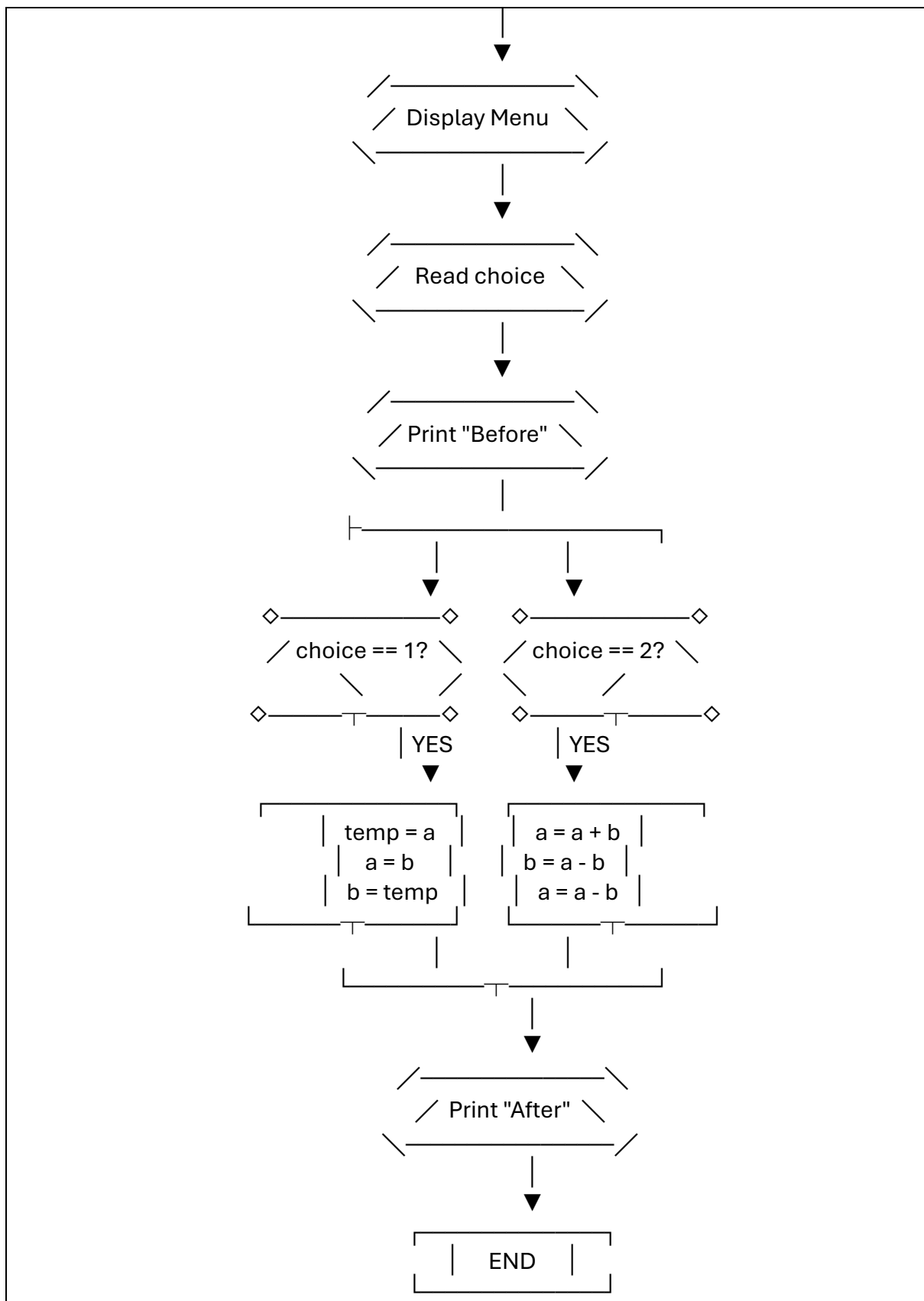
Question 1: Menu-Driven Program to Swap Two Numbers

1. Problem Statement Identification

- **Problem Statement:** Swap two input numbers using a third variable or without using a third variable based on the user's menu choice, displaying values before and after swapping.
- **Input:** Number 1 (a), Number 2 (b), Menu Choice (1 or 2).
- **Output:** Swapped values of a and b .
- **Formulas / Operations:**
 - **Option 1 (Using 3rd variable):** $\text{temp} = a, a = b, b = \text{temp}$
 - **Option 2 (Without 3rd variable):** $a = a + b, b = a - b, a = a - b$

2. Flowchart





3. pseudocode-

```

BEGIN
  DECLARE a, b, temp AS DOUBLE
  DECLARE choice AS INTEGER

  PRINT "Enter two numbers (a and b):"
  READ a, b

  PRINT "1. Swap using a third variable"
  PRINT "2. Swap without using a third variable"
  PRINT "Enter choice (1-2):"
  READ choice

  PRINT "Before Swapping: a = ", a, ", b = ", b

  IF choice == 1 THEN
    temp = a
    a = b
    b = temp
    PRINT "After Swapping: a = ", a, ", b = ", b
  ELSE IF choice == 2 THEN
    a = a + b
    b = a - b
    a = a - b
    PRINT "After Swapping: a = ", a, ", b = ", b
  ELSE
    PRINT "Invalid Choice"
  END IF
END

```

4. Algorithm

1. **Start**
2. Read numbers a and b.
3. Display menu options (1: Swap using 3rd variable, 2: Swap without 3rd variable).
4. Read choice.

5. Print values of a and b before swapping.

6. If choice == 1:

- Set temp = a, a = b, b = temp.

7. Else if choice == 2:

- Set a = a + b, b = a - b, a = a - b.

8. Else:

- Print "Invalid choice!".

9. Print swapped values of a and b.

10. **End**

5. Standard Test Case Table

Test Case	Input (a, b, choice)	Expected Output	Actual Output	Result (Pass/Fail)
1	10 20 1	Before: a=10, b=20 After: a=20, b=10	Before: a=10.00, b=20.00 After: a=20.00, b=10.00	Pass
2	10 20 2	Before: a=10,	Before: a=10.00,	Pass

Test Case	Input (a, b, choice)	Expected Output	Actual Output	Result (Pass/Fail)
		b=20 After: a=20, b=10	b=20.00 After: a=20.00, b=10.00	
3	-5 15 1	Before: a=-5, b=15 After: a=15, b=-5	Before: a=-5.00, b=15.00 After: a=15.00, b=-5.00	Pass
4	7 3 3	Invalid choice!	Invalid choice!	Pass

6. C Program

```
#include <stdio.h>

int main(void) {
    double a, b, temp;
    int choice;

    printf("Enter two numbers (a and b): ");
    if (scanf("%lf %lf", &a, &b) != 2) {
        printf("Invalid Input\n");
        return 1;
    }
}
```

```

printf("\n--- Swapping Menu ---\n");
printf("1. Swap using a third variable\n");
printf("2. Swap without using a third variable\n");
printf("Enter your choice (1 or 2): ");
if (scanf("%d", &choice) != 1) {
    printf("Invalid Input\n");
    return 1;
}

printf("\nBefore Swapping: a = %.2f, b = %.2f\n", a, b);

if (choice == 1) {
    temp = a;
    a = b;
    b = temp;
    printf("After Swapping (using 3rd variable): a = %.2f, b = %.2f\n", a, b);
} else if (choice == 2) {
    a = a + b;
    b = a - b;
    a = a - b;
    printf("After Swapping (without 3rd variable): a = %.2f, b = %.2f\n", a,
b);
} else {
    printf("Invalid choice!\n");
}

return 0;
}

```

7. Evaluation & Screenshots

```
Session Actions Edit View Help
kali@kali:~/HDOOC_Day9
q1.c
#include <stdio.h>

int main(void) {
    double a, b, temp;
    int choice;

    printf("Enter two numbers (a and b): ");
    if (scanf("%lf %lf", &a, &b) != 2) {
        printf("Invalid Input\n");
        return 1;
    }

    printf("\n--- Swapping Menu ---\n");
    printf("1. Swap using a third variable\n");
    printf("2. Swap without using a third variable\n");
    printf("Enter your choice (1 or 2): ");
    if (scanf("%d", &choice) != 1) {
        printf("Invalid Input\n");
        return 1;
    }

    printf("\nBefore Swapping: a = %.2f, b = %.2f\n", a, b);

    if (choice == 1) {
        temp = a;
        a = b;
        b = temp;
        printf("After Swapping (using 3rd variable): a = %.2f, b = %.2f\n", a, b);
    } else if (choice == 2) {
        a = a + b;
        b = a - b;
        a = a - b;
        printf("After Swapping (without 3rd variable): a = %.2f, b = %.2f\n", a, b);
    } else {
        printf("Invalid choice\n");
    }

    return 0;
}
```

```
kali@kali:~/HDOOC_Day9
$ cd HDOOC_Day9
$ nano q1.c
$ gcc q1.c -o q1
$ ./q1
Enter two numbers (a and b): 10 20
--- Swapping Menu ---
1. Swap using a third variable
2. Swap without using a third variable
Enter your choice (1 or 2): 1
Before Swapping: a = 10.00, b = 20.00
After Swapping (using 3rd variable): a = 20.00, b = 10.00
kali@kali:~/HDOOC_Day9
```

Question 2: Menu-Driven Sum of First n Natural Numbers & Sum of Squares

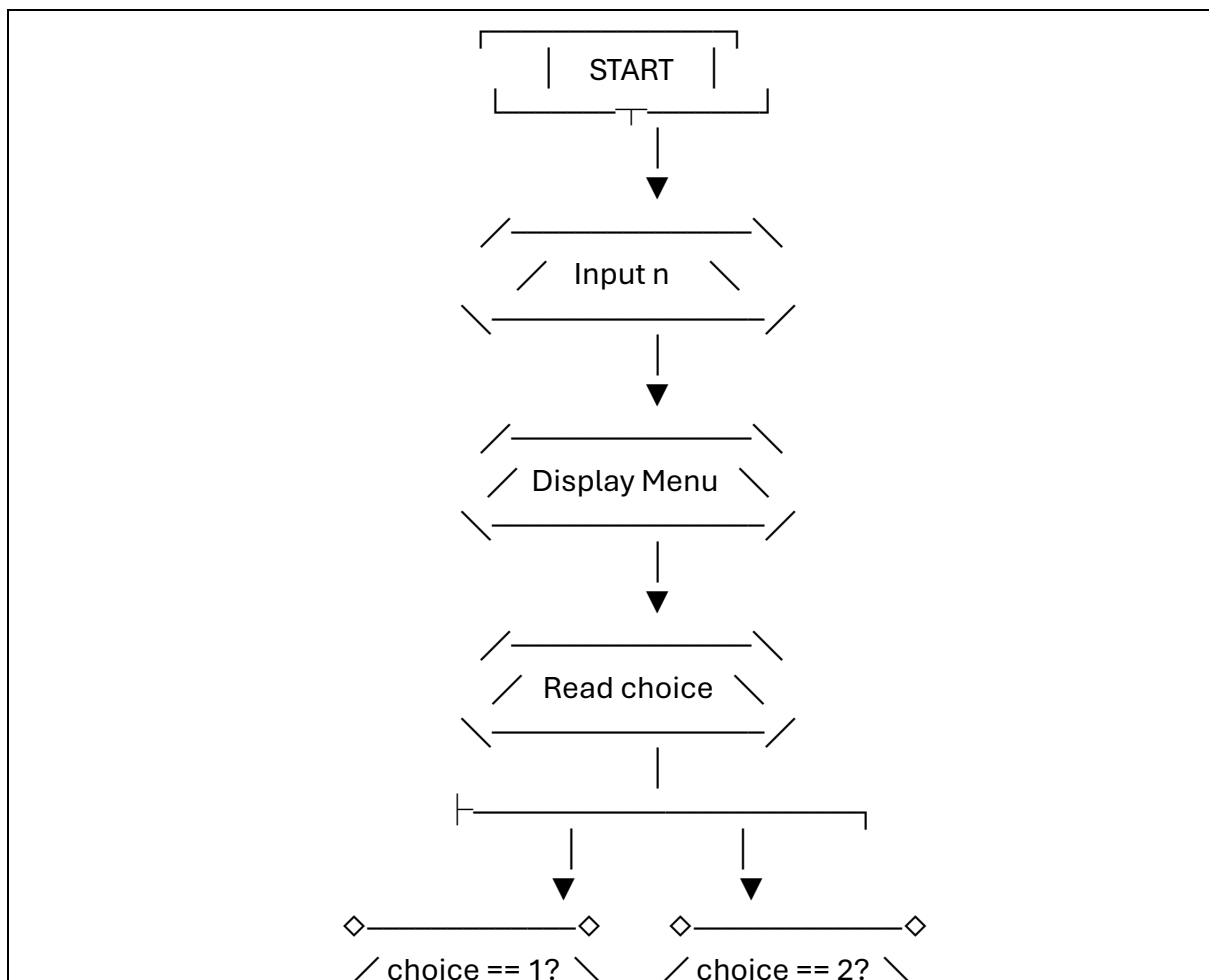
1. Problem Statement Identification

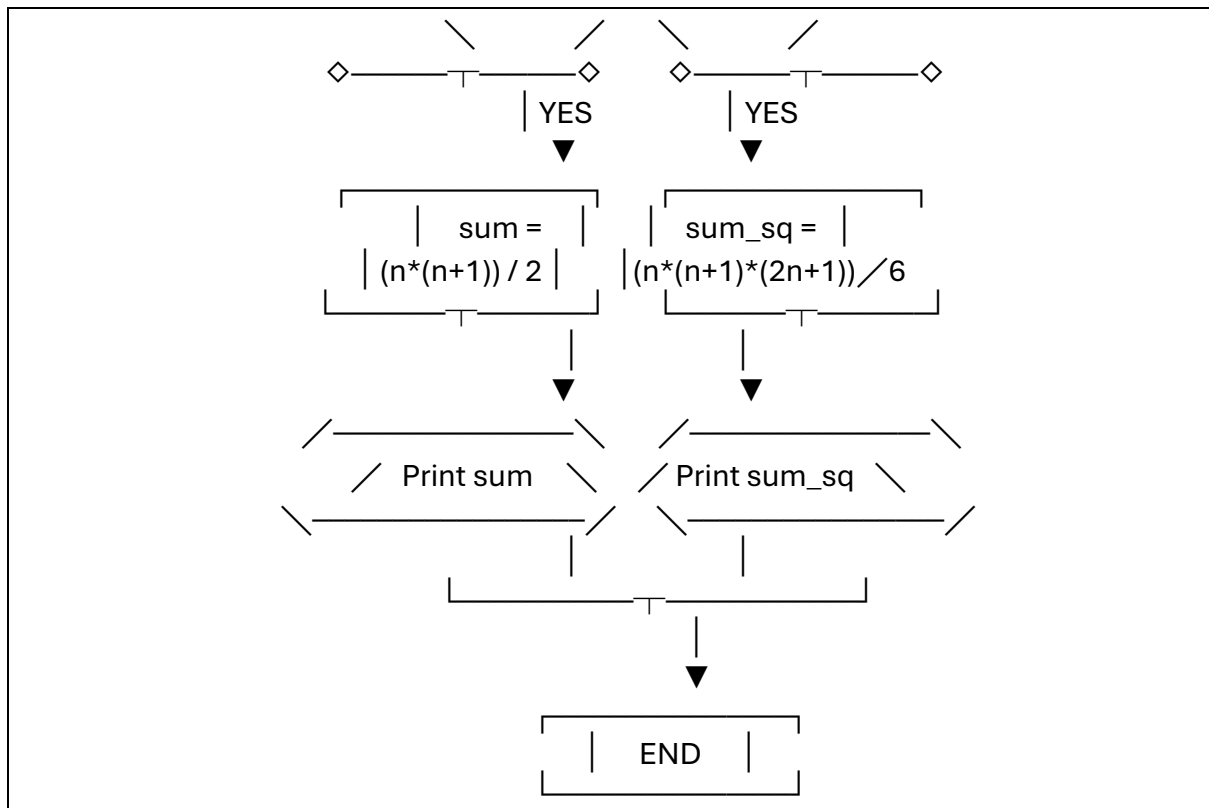
- **Problem Statement:** Calculate either the sum of the first n natural numbers or the sum of their squares based on user choice.
- **Input:** Positive integer (n), Menu Choice (1 or 2).
- **Output:** Calculated sum or sum of squares.
- **Formulas:**

$$\text{Sum} = \frac{n(n + 1)}{2}$$

$$\text{Sum of Squares} = \frac{n(n + 1)(2n + 1)}{6}$$

2. Flowchart





3.Pseudocode-

BEGIN

DECLARE n, sum, sum_sq AS LONG INTEGER

DECLARE choice AS INTEGER

PRINT "Enter a positive integer (n):"

READ n

PRINT "1. Sum of first n natural numbers"

PRINT "2. Sum of squares of first n natural numbers"

PRINT "Enter choice (1-2):"

READ choice

IF choice == 1 THEN

sum = (n * (n + 1)) / 2

PRINT "Sum: ", sum

ELSE IF choice == 2 THEN

sum_sq = (n * (n + 1) * (2 * n + 1)) / 6

PRINT "Sum of Squares: ", sum_sq

ELSE

<pre>PRINT "Invalid Choice" END IF END</pre>
--

4. Algorithm

1. Start

2. Read positive integer n.

**3. Display menu options (1: Sum of natural numbers,
2: Sum of squares).**

4. Read choice.

5. If choice == 1:

- **Calculate $\text{sum} = (n * (n + 1)) / 2$.**
- **Print sum.**

6. Else if choice == 2:

- **Calculate $\text{sum_sq} = (n * (n + 1) * (2 * n + 1)) / 6$.**
- **Print sum_sq.**

7. Else:

- **Print "Invalid choice!".**

8. End

5. Standard Test Case Table

Test Case	Input (n, choice)	Expected Output	Actual Output	Result (Pass/Fail)
1	10 1	Sum: 55	Sum of first 10 natural numbers: 55	Pass
2	10 2	Sum of Squares: 385	Sum of squares of first 10 natural numbers: 385	Pass
3	5 1	Sum: 15	Sum of first 5 natural numbers: 15	Pass
4	5 2	Sum of Squares: 55	Sum of squares of first 5 natural	Pass

Test Case	Input (n, choice)	Expected Output	Actual Output	Result (Pass/Fail)
			numbers: 55	

6. C Program

```
#include <stdio.h>

int main(void) {
    long long n;
    int choice;

    printf("Enter a positive integer (n): ");
    if (scanf("%lld", &n) != 1 || n <= 0) {
        printf("Invalid Input: n must be a positive integer.\n");
        return 1;
    }

    printf("\n--- Sum Menu ---\n");
    printf("1. Sum of first n natural numbers\n");
    printf("2. Sum of squares of first n natural numbers\n");
    printf("Enter your choice (1 or 2): ");
    if (scanf("%d", &choice) != 1) {
        printf("Invalid Input\n");
        return 1;
    }

    if (choice == 1) {
        long long sum = (n * (n + 1)) / 2;
        printf("Sum of first %lld natural numbers: %lld\n", n, sum);
    } else if (choice == 2) {
        long long sum_sq = (n * (n + 1) * (2 * n + 1)) / 6;
        printf("Sum of squares of first %lld natural numbers: %lld\n", n,
sum_sq);
    }
}
```

```

    } else {
        printf("Invalid choice!\n");
    }

    return 0;
}

```

7. Evaluation & Screenshots

```

GNU nano 2.8.0 q2.c
#include <stdio.h>

int main(void) {
    long long n;
    int choice;

    printf("Enter a positive integer (n): ");
    if (scanf("%lld", &n) != 1 || n < 0) {
        printf("Invalid Input: n must be a positive integer.\n");
        return 1;
    }

    printf("\n--- Sum Menu ---\n");
    printf("1. Sum of first n natural numbers\n");
    printf("2. Sum of squares of first n natural numbers\n");
    printf("Enter your choice (1 or 2): ");
    if (scanf("%d", &choice) != 1) {
        printf("Invalid Input\n");
        return 1;
    }

    if (choice == 1) {
        long long sum = (n * (n + 1)) / 2;
        printf("Sum of first %lld natural numbers: %lld\n", n, sum);
    } else if (choice == 2) {
        long long sum_sq = (n * (n + 1) * (2 * n + 1)) / 6;
        printf("Sum of squares of first %lld natural numbers: %lld\n", n, sum_sq);
    } else {
        printf("Invalid choice!\n");
    }

    return 0;
}

```

```

(kali㉿kali)-[~/HDOC_Day9]
$ nano q2.c

(kali㉿kali)-[~/HDOC_Day9]
$ gcc q2.c -o q2
./q2
Enter a positive integer (n): 10

--- Sum Menu ---
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
Enter your choice (1 or 2): 1
Sum of first 10 natural numbers: 55

(kali㉿kali)-[~/HDOC_Day9]
$

```

Question 3: Menu-Driven Simple Interest & Compound Interest Calculation

1. Problem Statement Identification

- **Problem Statement:** Compute simple interest with total amount payable or compound interest with compound amount given principal, rate, and time based on user menu choice.
- **Input:** Principal (P), Rate of interest (R), Time (T), Menu Choice (1 or 2).
- **Output:** Interest amounts and Total Payable Amount.
- **Formulas:**

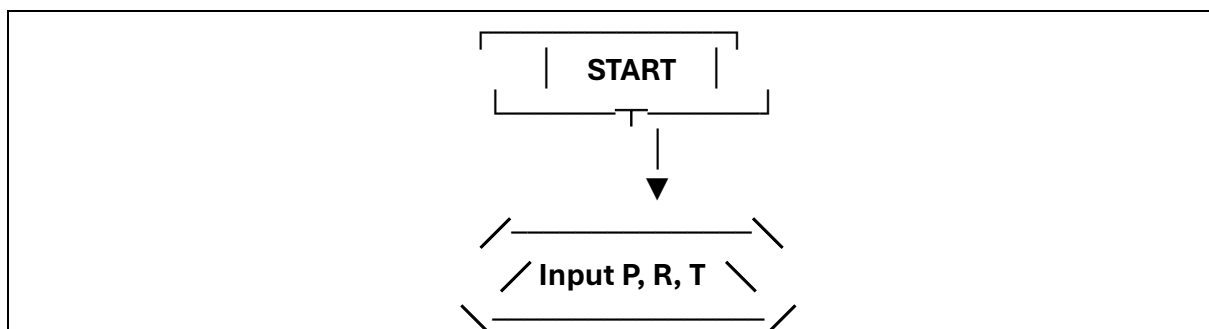
$$SI = \frac{P \cdot R \cdot T}{100}$$

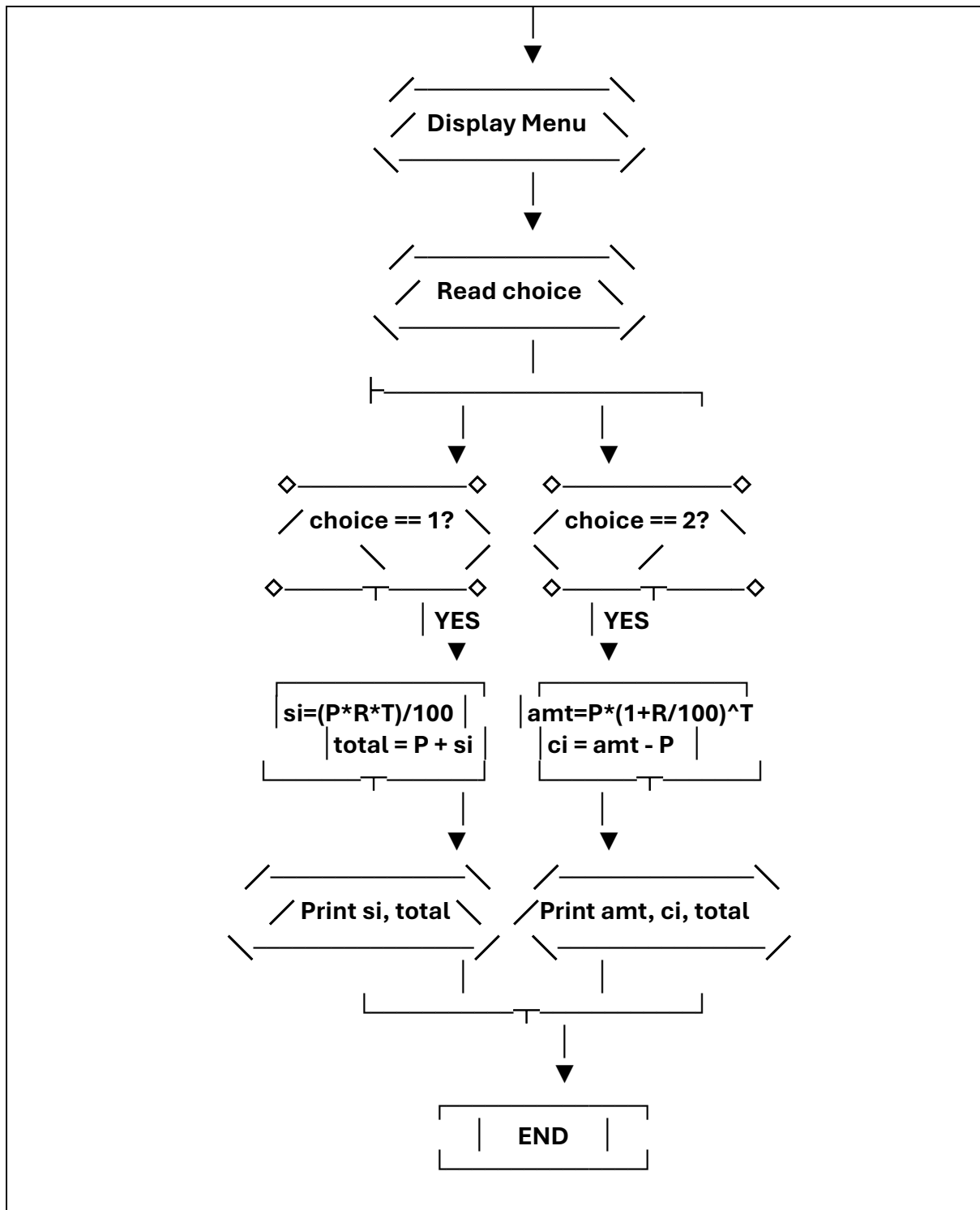
$$\text{Total Amount (SI)} = P + SI$$

$$\text{Compound Amount} = P \cdot \left(1 + \frac{R}{100}\right)^T$$

$$CI = \text{Compound Amount} - P$$

2. Flowchart





3.Psuedocode-

```

BEGIN
  DECLARE principal, rate, time, si, ci, compound_amount,
total_amount AS DOUBLE
  DECLARE choice AS INTEGER

```

```
PRINT "Enter Principal, Rate (%), and Time (years):"
READ principal, rate, time

PRINT "1. Simple Interest & Total Amount"
PRINT "2. Compound Interest & Compound Amount"
PRINT "Enter choice (1-2):"
READ choice

IF choice == 1 THEN
    si = (principal * rate * time) / 100.0
    total_amount = principal + si
    PRINT "Simple Interest: ", si
    PRINT "Total Payable Amount: ", total_amount
ELSE IF choice == 2 THEN
    compound_amount = principal * POWER((1.0 + rate / 100.0), time)
    ci = compound_amount - principal
    PRINT "Compound Amount: ", compound_amount
    PRINT "Compound Interest: ", ci
    PRINT "Total Payable Amount: ", compound_amount
ELSE
    PRINT "Invalid Choice"
END IF
END
```

4. Algorithm

1. Start

2. Read principal, rate, and time.

3. Display menu options (1: Simple Interest, 2: Compound Interest).

4. Read choice.

5. If choice == 1:

- Compute $si = (\text{principal} * \text{rate} * \text{time}) / 100.0$.

- **Compute total_amount = principal + si.**
- **Print si and total_amount.**

6. Else if choice == 2:

- **Compute compound_amount = principal * pow((1.0 + rate / 100.0), time).**
- **Compute ci = compound_amount - principal.**
- **Print compound_amount, ci, and total payable amount.**

7. Else:

- **Print "Invalid choice!".**

8. End

5. Standard Test Case Table

Test Case	Input (P, R, T, choice)	Expected Output	Actual Output	Result (Pass/Fail)
1	1000 5 2 1	SI: 100.00, Total: 1100.00	Simple Interest: 100.00 Total Payable	Pass

Test Case	Input (P, R, T, choice)	Expected Output	Actual Output	Result (Pass/Fail)
			Amount: 1100.00	
2	1000 5 2 2	Compound Amount: 1102.50, CI: 102.50	Compound Amount: 1102.50 Compound Interest: 102.50	Pass
3	5000 10 3 1	SI: 1500.00, Total: 6500.00	Simple Interest: 1500.00 Total Payable Amount: 6500.00	Pass
4	5000 10 3 2	Compound Amount: 6655.00,	Compound Amount: 6655.00	Pass

Test Case	Input (P, R, T, choice)	Expected Output	Actual Output	Result (Pass/Fail)
		CI: 1655.00	Compound Interest: 1655.00	

6. C Program

```

#include <stdio.h>
#include <math.h>

int main(void) {
    double principal, rate, time;
    int choice;

    printf("Enter Principal, Rate of Interest (%%), and Time (years): ");
    if (scanf("%lf %lf %lf", &principal, &rate, &time) != 3 || principal <= 0
|| rate <= 0 || time <= 0) {
        printf("Invalid Input: All values must be positive.\n");
        return 1;
    }

    printf("\n--- Interest Calculation Menu ---\n");
    printf("1. Calculate Simple Interest & Total Amount Payable\n");
    printf("2. Calculate Compound Interest & Compound Amount\n");
    printf("Enter your choice (1 or 2): ");
    if (scanf("%d", &choice) != 1) {
        printf("Invalid Input\n");
        return 1;
    }

    if (choice == 1) {

```

```

double si = (principal * rate * time) / 100.0;
double total_amount = principal + si;
printf("Simple Interest: %.2f\n", si);
printf("Total Payable Amount: %.2f\n", total_amount);
} else if (choice == 2) {
    double compound_amount = principal * pow((1.0 + rate / 100.0),
time);
    double ci = compound_amount - principal;
    printf("Compound Amount: %.2f\n", compound_amount);
    printf("Compound Interest: %.2f\n", ci);
    printf("Total Payable Amount: %.2f\n", compound_amount);
} else {
    printf("Invalid choice!\n");
}

return 0;
}

```

7. Evaluation & Screenshots

The screenshot shows a terminal window with the GNU nano 2.8.0 editor. The code is a C program for calculating simple and compound interest. The program prompts the user to enter principal, rate, and time, then asks for a choice (1 for simple interest, 2 for compound interest). It calculates the interest and total payable amount based on the choice and prints the results. The terminal output shows the program running successfully and calculating the interest for a given input.

```

GNU nano 2.8.0
#include <stdio.h>
#include <math.h>

int main(void) {
    double principal, rate, time;
    int choice;

    printf("Enter Principal, Rate of Interest (%), and Time (years): ");
    if (scanf("%lf %lf %lf", &principal, &rate, &time) != 3 || principal < 0 || rate < 0 || time < 0) {
        printf("Invalid Input: All values must be positive.\n");
        return 1;
    }

    printf("\n--- Interest Calculation Menu ---\n");
    printf("1. Calculate Simple Interest & Total Amount Payable\n");
    printf("2. Calculate Compound Interest & Compound Amount\n");
    printf("Enter your choice (1 or 2): ");
    if (scanf("%d", &choice) != 1) {
        printf("Invalid Input\n");
        return 1;
    }

    if (choice == 1) {
        double si = (principal * rate * time) / 100.0;
        double total_amount = principal + si;
        printf("Simple Interest: %.2f\n", si);
        printf("Total Payable Amount: %.2f\n", total_amount);
    } else if (choice == 2) {
        double compound_amount = principal * pow((1.0 + rate / 100.0), time);
        double ci = compound_amount - principal;
        printf("Compound Amount: %.2f\n", compound_amount);
        printf("Compound Interest: %.2f\n", ci);
        printf("Total Payable Amount: %.2f\n", compound_amount);
    } else {
        printf("Invalid choice!\n");
    }

    return 0;
}

```

```
(kali㉿kali)-[~/HDOC_Day9]
$ nano q3.c

(kali㉿kali)-[~/HDOC_Day9]
$ gcc q3.c -o q3 -lm
./q3
Enter Principal, Rate of Interest (%), and Time (years): 1000 5 2

— Interest Calculation Menu —
1. Calculate Simple Interest & Total Amount Payable
2. Calculate Compound Interest & Compound Amount
Enter your choice (1 or 2): 1
Simple Interest: 100.00
Total Payable Amount: 1100.00

(kali㉿kali)-[~/HDOC_Day9]
$
```